LINEAR ALGEBRA · LESSON 6

Column Space and Nullspace

Two questions every matrix answers: **which targets can I reach?** and **which trades change nothing?**

With portfolios, hedges and put–call parity.

C(A) and N(A) | $x = x_p + x_n$ | MIT 18.06, Lecture 6

THE SETUP

Funds as columns

Rows: budget (₹ lakh), return (₹ thousand), exposure. Columns: Nifty, bonds, and a **balanced** fund (half Nifty, half bonds).

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix}$$

	Column space $C(A)$	**Nullspace** $N(A)$
Question	Which targets can I reach?	Which trades change nothing?
Made of	targets b	trades x

Ax = b is solvable exactly when b ∈ C(A)

Ax is always a mix of the columns:

$$A\begin{bmatrix}7\\3\\0\end{bmatrix} = 7\begin{bmatrix}1\\12\\1\end{bmatrix} + 3\begin{bmatrix}1\\7\\0\end{bmatrix} + 0\begin{bmatrix}1\\9.5\\0.5\end{bmatrix} = \begin{bmatrix}10\\105\\7\end{bmatrix}$$

So a target can be hit only if it's a mix of the columns. Balanced = ½ Nifty + ½ bonds adds no new direction: $C(A)$ stays the plane $R = 7B + 5E$.

Twin index funds

Two Nifty funds from different houses: both columns $(1, 12)$.

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 12-12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Move ₹1 lakh from one twin to the other: **nothing changes.** All multiples $c(1, -1)$ form the nullspace, a line.

Find it by elimination

$$U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$$

No pivot in column 3, so balanced is **free**. Choose $z = -2$: then $y = 1$, $x = 1$.

Check: $1 + 1 - 2 = 0,\ \ 12 + 7 - 19 = 0,\ \ 1 + 0 - 1 = 0$ ✓

MEANING

Buy ₹1L Nifty + ₹1L bonds, sell ₹2L balanced: you built the balanced fund yourself.

A SUBSPACE

Do-nothing + do-nothing = do-nothing

If $Av = 0$ and $Aw = 0$:

$$A(v + w) = 0 + 0 = 0 \qquad A(cv) = c \cdot 0 = 0$$

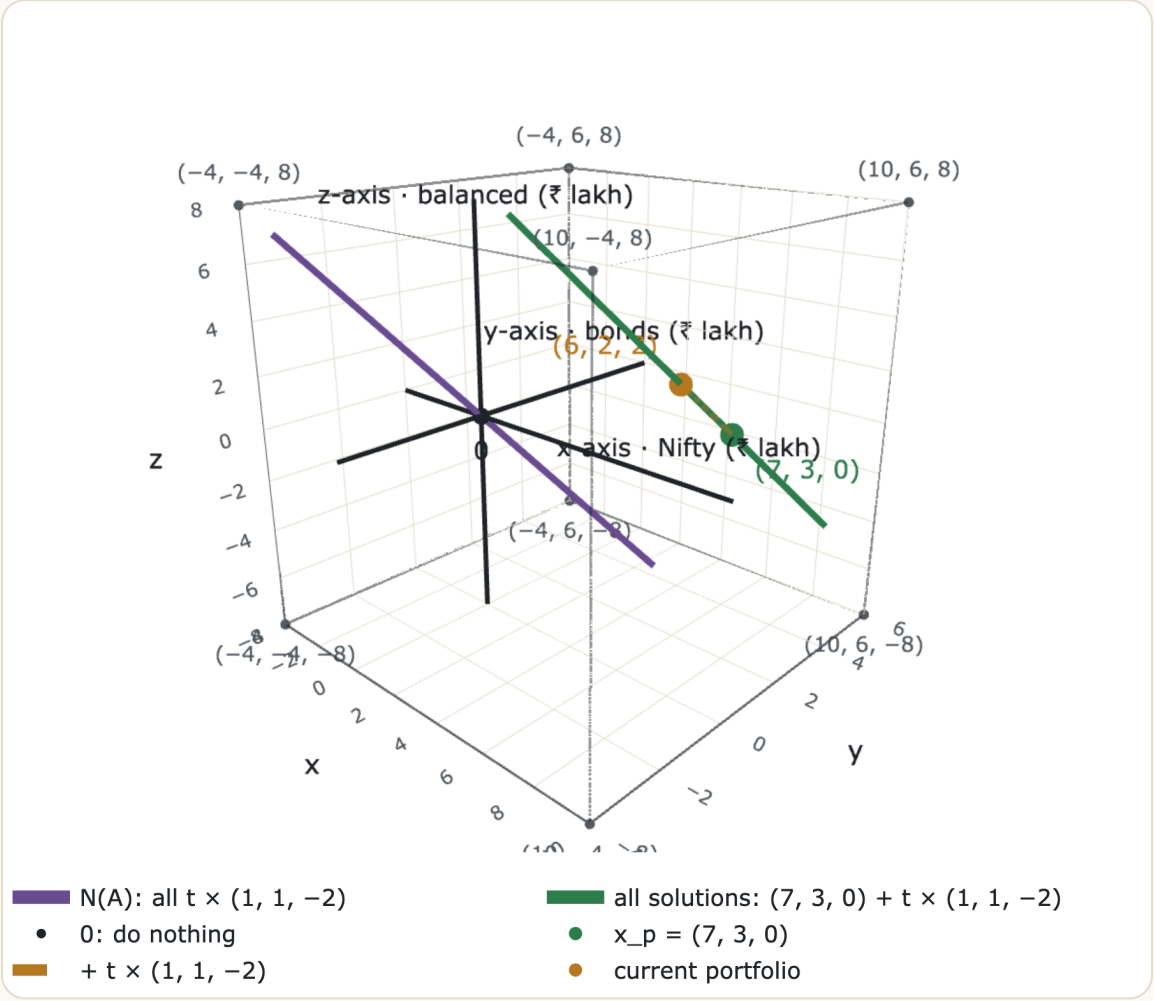
So $N(A)$ is always a subspace, of the **trade** space $\mathbb{R}^n$. $C(A)$ is a subspace of the **target** space $\mathbb{R}^m$.

Swap balanced for midcap and elimination finds 3 pivots: $N(A) = \{0\}$, and every target has **exactly one** portfolio.

EVERY SOLUTION

One answer + the nullspace

$$x = \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$



$(7, 3, 0)$, $(6, 2, 2)$, $(4, 0, 6)$ all give ₹10L, ₹105K, exposure 7.

NOT A SUBSPACE

The solutions are a shifted line

$(0, 0, 0)$ isn't a solution: doing nothing has budget 0, not 10.

Add two solutions: $(7, 3, 0) + (6, 2, 2) = (13, 5, 2)$ hits $(20, 210, 14)$, which is $2b$, not b.

> RULE
> Linear rules "$= 0$" give a subspace (a nullspace). A non-zero right side shifts it off the origin.

IN MARKETS

Put–call parity is a nullspace

Payoff per lot at 19,800 / 20,000 / 20,200: futures $(-200, 0, 200)$, 20,000 call $(0, 0, 200)$, 20,000 put $(200, 0, 0)$.

$$1 \cdot \text{fut} - 1 \cdot \text{call} + 1 \cdot \text{put} = (0, 0, 0)$$

So "1 future + 2 calls" and "3 futures + 2 puts" pay the same everywhere. If one costs more, sell it and buy the other.

RECAP

Key takeaways

1. $Ax = b$ solvable $\Leftrightarrow$ $b \in C(A)$.
2. $N(A)$ = all x with $Ax = 0$: do-nothing trades.
3. Free columns in elimination $\Rightarrow$ non-zero nullspace.
4. $N(A) = \{0\}$ $\Rightarrow$ one portfolio per target.
5. All solutions = x_p + nullspace.
6. Solutions of $Ax = b \neq 0$ are not a subspace.

QUANT FROM FIRST PRINCIPLES

The full interactive lesson is free

A nullspace lab for your own funds and targets, a slider through every solution, and worked hedging and options examples:

learn.krisarj.com

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